

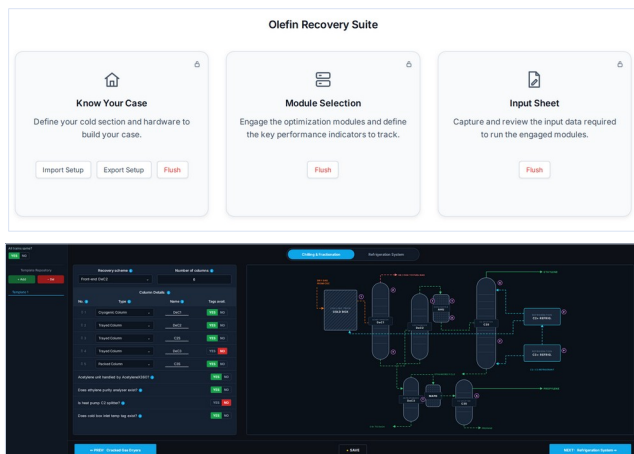
PRODUCT OVERVIEW

RecoveryX360 is a decision-support application for the olefin cold section. Domain and ML agents are configured on the unit's own historian and sensor tag data, covering the chilling train / cold box, demethanizer, deethanizer, C2 splitter, depropanizer, MAPD converter, C3 splitter and the ethylene / propylene refrigeration systems (acetylene removal is covered by AcetyleneX360). It tracks ethylene losses to fuel gas, product purity giveaway, column hydraulics, cold box health and refrigeration power. Output reaches the process engineer as ranked, quantified actions, each reported against the parameter that caused it. Every model agent is configured, validated and commissioned per site by Ingenero process engineers, so the deployment reflects that unit's configuration and operating envelope.



01 Unit Configuration

Shown: Olefin recovery suite - interactive configuration, modules and sensor entry

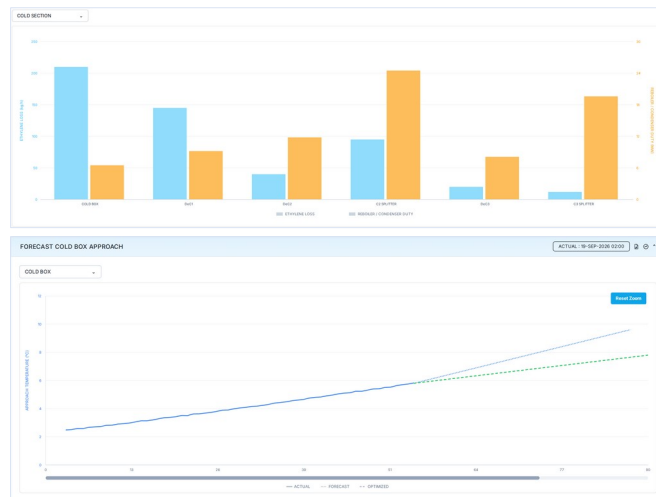


- Know Your Case** Configure recovery scheme, cold box, columns and refrigeration
- Module Selection** After configuring the cold section, select the modules - Domain & AI agents
- Input Sheet** Based on configuration & Module selection - relevant sensor info needs to be filled
- Interactive Config** Chilling train, columns & refrigeration interactive PFD available



02 Plant Data and Model Execution

Shown: column-wise ethylene loss & duty, cold box approach forecast vs optimum.



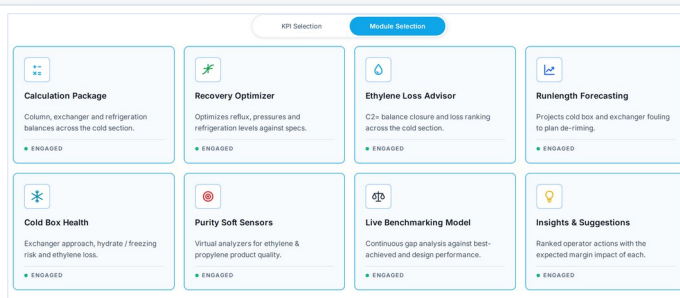
- Inputs** Sensor tag entry with design tag values and online analyser data
- Processing** tag validation, steady state framing and column regime detection
- Outputs** predicted KPIs, ranked actionable with cause, trends, alerts



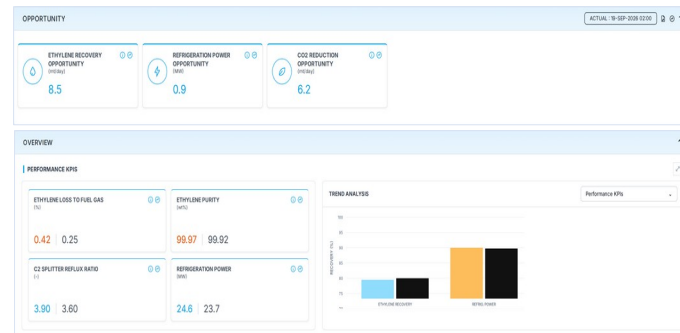
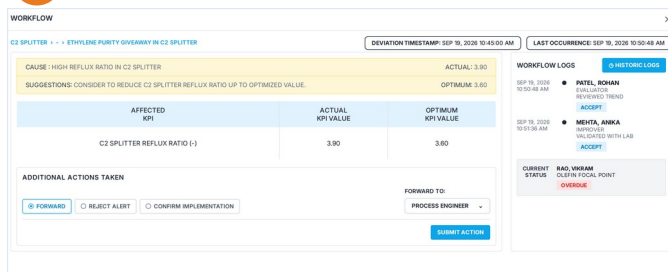
03 AI-Powered Model Agent Suite

Agent/Module selection shared along with detailing of the recovery specific details

- Chilling Train & Cold Box** Tracks exchanger approach and ethylene loss to methane / fuel gas.
- Column Optimization** Reflux, pressure and duty optimization for DeC1, DeC2, C2 & C3 splitters.
- Ethylene Loss Tracking** Closes C2+ balance and ranks losses to fuel gas, ethane and C3+.
- Product Purity Giveaway** Soft sensors for ethylene / propylene purity vs spec giveaway.
- Refrigeration Systems** C2 / C3 refrigeration compressor power and level optimization.
- Column Hydraulics** Flooding / weeping margin estimation and early warnings.
- Cold Box Fouling Forecast** Forecasts exchanger approach rise to plan de-riming and warm-up



04 Opportunity and Operator Actions



+0.4%

Ethylene recovery
reduced losses in cold box & demethanizer

3-5%

Refrigeration power
C2 / C3 refrigeration optimization

20-30%

Purity giveaway
soft sensor based spec targeting

Basis: optimization respects cold section, CGC discharge and product spec limits; suggestions take care of step change for smooth implementation.



05 Platform

Historian connectivity

plug and play with leading plant historians

Simulation integration

third party simulators integrate without custom work

Daily reporting

downloadable daily reports with interactive trends

Gen AI assistant

answers plant questions in the operating workflow

Build your own dashboard

engineers build the views their unit needs

DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

Front-end DeC1 or DeC2 schemes; selectable KPIs; virtual analyzers

SCALE

Repeat sites reuse the configuration, with no/minimal rework per plant.

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